4.4 ASA and AAS Congruence

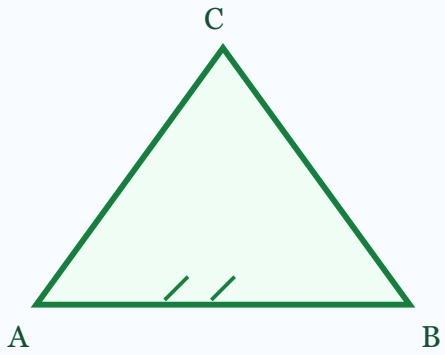
The **ASA Postulate** proves triangles congruent from two pairs of congruent angles plus the *included* side (the side between those two angles). The **AAS Theorem** proves congruence from two pairs of congruent angles plus a non-included side — a side that lies opposite one of the given angles rather than between them. AAS actually reduces to ASA: since two angles are congruent, the Angle-Sum Theorem forces the third pair of angles to be congruent too, which makes the given side included relative to that third angle. Note that **AAA is never a valid congruence criterion** — matching all three angles only guarantees the triangles are the same shape (similar), not the same size. This worksheet applies ASA and AAS with numeric, algebraic, and overlapping-triangle reasoning, and contrasts them with AAA.

Examples

Example 1: Apply ASA

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $AB = DE$ (the side between $\angle A$ and $\angle B$), and $\angle B \cong \angle E$. What can you conclude?

Solution



Two angles and the side directly between them are marked congruent.

Identify the given parts: $\angle A \cong \angle D$, $\angle B \cong \angle E$, and side $\overline{AB}$ lies between $\angle A$ and $\angle B$ — it is the included side.

Two angles and the included side are congruent.

Conclusion: $\triangle ABC \cong \triangle DEF$ by ASA.

Example 2: Apply AAS

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $BC = EF$. Is $\overline{BC}$ included between the two given angles? What can you conclude?

Solution

Side $\overline{AB}$ (not $\overline{BC}$) lies between $\angle A$ and $\angle B$, so $\overline{BC}$ is *not* included — it lies opposite $\angle A$.

We have two pairs of congruent angles and one pair of congruent non-included sides.

Conclusion: $\triangle ABC \cong \triangle DEF$ by AAS.

Example 3: Why AAS actually works

Explain why AAS guarantees congruence, using the Angle-Sum Theorem.

Solution

Given $\angle A \cong \angle D$ and $\angle B \cong \angle E$, apply the Angle-Sum Theorem to both triangles:

$\angle C = 180 - \angle A - \angle B$ and $\angle F = 180 - \angle D - \angle E$.

Since $\angle A = \angle D$ and $\angle B = \angle E$, the right-hand sides are equal, so $\angle C \cong \angle F$ as well.

Now side $\overline{BC}$ is included between $\angle B$ and the newly-shown $\angle C$ (and likewise $\overline{EF}$ between $\angle E$ and $\angle F$).

Conclusion: AAS is really ASA in disguise — the third angle pair is congruent "for free" from the Angle-Sum Theorem, making the given side included after all.

Example 4: Algebra for ASA

In $\triangle ABC$ and $\triangle DEF$, $\angle A = 3x + 5$, $\angle D = 2x + 25$ (given congruent), $\angle B \cong \angle E$, and included side $AB = DE = 14$. Find x and $\angle A$.

Solution

Set the two angle expressions equal (since $\angle A \cong \angle D$):

$3x + 5 = 2x + 25$

$x = 20$.

$\angle A = 3(20) + 5 = 65°$ (check: $\angle D = 2(20) + 25 = 65°$ ✓).

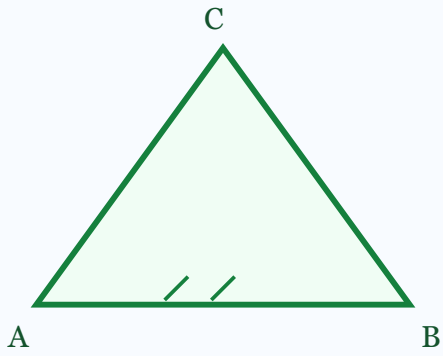
With $\angle A \cong \angle D$, $\angle B \cong \angle E$, and included side $AB = DE = 14$:

Conclusion: $x = 20$; $\triangle ABC \cong \triangle DEF$ by ASA.

Example 5: Shared side via angle bisector (AAS)

$\overline{BD}$ bisects $\angle ABC$ (so $\angle ABD \cong \angle CBD$), and $\angle A \cong \angle C$. Prove $\triangle ABD \cong \triangle CBD$.

Solution



$\overline{BD}$ splits the triangle; the shared side BD is opposite the marked angles at A and C.

Identify the parts: $\angle A \cong \angle C$ (given) and $\angle ABD \cong \angle CBD$ (bisector, given) — two pairs of angles.

$\overline{BD} \cong \overline{BD}$ by the Reflexive Property (shared side).

In each triangle, $\overline{BD}$ is opposite $\angle A$ (or $\angle C$) — it is not between the two given angles, so it is a non-included side.

Conclusion: $\triangle ABD \cong \triangle CBD$ by AAS.

Example 6: AAA is not enough

$\triangle ABC$ has angles 40°, 60°, 80° and sides 5, 7, 8. $\triangle DEF$ has the same three angles 40°, 60°, 80° but sides 10, 14, 16. Are the triangles congruent?

Solution

Compare the angles: all three pairs match exactly ($40° = 40°$, $60° = 60°$, $80° = 80°$) — this is AAA.

Compare the sides: $5 \nmid 10$, $7 \nmid 14$, $8 \nmid 16$ — none of the corresponding sides match.

The triangles are the same shape (each side of $\triangle DEF$ is exactly double the corresponding side of $\triangle ABC$) but not the same size.

Conclusion: Not congruent — matching angles only (AAA) proves the triangles are *similar*, never congruence on its own.

Example 7: CPCTC after ASA

$\triangle ABC \cong \triangle DEF$ by ASA, with $AB = 10$, $BC = 14$, and $\angle B = 72°$. Find DE, EF, and $\angle E$.

Solution

The correspondence from $\triangle ABC \cong \triangle DEF$ is $A \leftrightarrow D, B \leftrightarrow E, C \leftrightarrow F$.

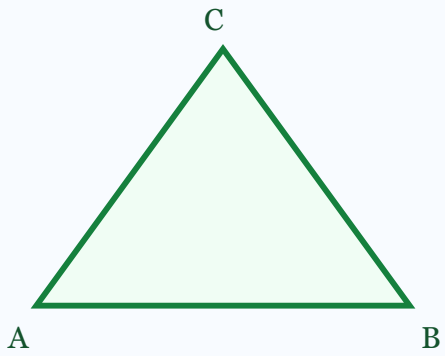
By CPCTC: $DE = AB = 10$, $EF = BC = 14$, and $\angle E = \angle B = 72°$.

Answer: $DE = 10$, $EF = 14$, $\angle E = 72°$.

Example 8: Roof brackets (ASA in context)

Two triangular roof brackets each have base angles $55°$ and $80°$ with the included side (between them) measuring 12 in. Are the brackets congruent? Find the third angle of each.

Solution



Each bracket is a triangle with two known base angles and the side between them.

Both brackets have $\angle 1 = 55°$, $\angle 2 = 80°$, and the included side $= 12$ in — matching ASA information.

Find the third angle using the Angle-Sum Theorem: $180 - 55 - 80 = 45°$.

Conclusion: the brackets are congruent by ASA; each has a third angle of $45°$.

Example 9: Algebra with two given angle pairs

In $\triangle ABC$ and $\triangle DEF$, $\angle A = (2x + 6)°$ and $\angle D = 50°$ are congruent; $\angle B = 65° = \angle E$; and $AB = DE = 9$ is the side between $\angle A$ and $\angle B$. Find x and name the postulate that proves the triangles congruent.

Solution

Set $\angle A = \angle D$:

$2x + 6 = 50$

$2x = 44 \Rightarrow x = 22.$

So $\angle A = 50°$, matching $\angle D$; also $\angle B = \angle E = 65°$ (already given).

Side $\overline{AB}$ is included between $\angle A$ and $\angle B$ in $\triangle ABC$ (and $\overline{DE}$ between $\angle D$ and $\angle E$ in $\triangle DEF$).

Conclusion: $x = 22$; $\triangle ABC \cong \triangle DEF$ by ASA.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: ASA meaning

What does ASA stand for?

Answer: **Verification:**

Problem 2: AAS meaning

What does AAS stand for?

Answer: **Verification:**

Problem 3: Included side

In ASA, where must the given side be located?

Answer: **Verification:**

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Problem 4: Conclude ASA

Two angles and the included side are congruent between two triangles. Which postulate applies?

Answer: **Verification:**

Problem 5: Conclude AAS

Two angles and a non-included side are congruent. Which postulate applies?

Answer: **Verification:**

Problem 6: AAA?

Is matching all three pairs of angles (AAA) enough to prove congruence?

Answer: **Verification:**

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Problem 7: Third angle

If two pairs of angles are congruent, why must the third pair also be congruent?

Answer: **Verification:**

Problem 8: Shared angle

Two triangles share an angle at a common vertex. Which property justifies the angle is congruent to itself?

Answer: **Verification:**

Problem 9: Correspondence

$\triangle ABC \cong \triangle XYZ$ by ASA. $\angle B$ corresponds to which angle?

Answer: **Verification:**

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Problem 10: Choose postulate

Marks show: angle, included side, angle. Which postulate applies?

Answer: **Verification:**

Problem 11: Choose postulate

Marks show: angle, angle, side opposite one of them. Which postulate applies?

Answer: **Verification:**

Problem 12: After proof

After proving triangles congruent, corresponding sides are congruent by what idea?

Answer: **Verification:**

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Problem 13: Challenge — algebra + CPCTC

In $\triangle ABC$ and $\triangle DEF$, $\angle A = (4x - 8)°$ and $\angle D = 60°$ are congruent; $\angle B = 45° = \angle E$; and $AB = DE = 13$ is the included side. Find x, name the postulate, then find $\angle C$ and $\angle F$.

Answer: **Verification:**

Problem 14: Challenge — AAS with algebra

Two triangular braces each have angles $50°$ and $70°$. The side opposite the $70°$ angle measures $3x - 2$ in one brace and 16 in the other, and the braces are congruent by AAS. Find x and the third angle of each brace.

Answer: **Verification:**

Problem 15: Challenge — bisector proof with algebra

$\overline{BD}$ bisects $\angle ABC$, with $\angle ABD = (3x + 5)°$ and $\angle CBD = (2x + 15)°$. Also $\angle A \cong \angle C = 48°$, and $\overline{BD}$ is shared (opposite the $48°$ angles). Find x, name the postulate proving $\triangle ABD \cong \triangle CBD$, and find $\angle ADB$.

Answer: **Verification:**

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✓ Answer Key

1. Angle-Side-Angle
2. Angle-Angle-Side
3. Between the two given angles (included)
4. ASA
5. AAS
6. No — AAA proves similarity, not congruence
7. By the Angle-Sum Theorem, the third angle equals $180°$ minus the other two, which are already equal
8. Reflexive Property
9. $\angle Y$
10. ASA
11. AAS
12. CPCTC
13. $x = 17$; ASA; $\angle C = \angle F = 75°$
14. $x = 6$; third angle $= 60°$ in each
15. $x = 10$; $\angle ABD = \angle CBD = 35°$; AAS; $\angle ADB = 97°$

Great work — keep practising!